Roll No.: DL AI U1AID24135

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Amrita School of Artificial Intelligence, Faridabad

B.Tech Mid-Term Examinations – October 2025

Third Semester

Artificial Intelligence & Data Science

23AID203 Software-Defined Communication Systems

Duration: Two hours | Maximum: 50 Marks

Course Outcomes (COs):

CO1	Understand and analyze the fundamentals of communication systems, signals, LTI systems, and modulation basics; simulate them using MATLAB/Simulink.
CO2	Apply concepts of analog modulation (AM, FM, PM) and demodulation, and evaluate their performance using MATLAB/Simulink.
CO3	Analyze and implement digital modulation/demodulation techniques (ASK, FSK, PSK, QPSK, OFDM, MIMO) and explore AI applications in communication systems.
CO4	Demonstrate practical understanding of Software Defined Radio (SDR) platforms for analog/digital modulation schemes, simulation, signal analysis, and spectrum visualization using MATLAB, GRC, RTL-SDR, and Adalm Pluto.

Answer all questions

Question No-1

Q. 1A. Check that the system described by y(t)= t.x(t)+1 is time variant or invariant system.? [2][CO1][BTL1]

Q. 1B. Determine the periodicity of the signal x(t)=2cos(t+3) + 3cos(t/3). [2][CO1][BTL1]

Q. 1C. Find the even and odd components of x(t) = $e^{-t/4}$.u(t). [2][CO1][BTL1]

Q. 1D. Sketch the following signal $x(t)$ and represent in terms of singularity function.

[2][CO1][BTL2]

$$x(t) = \begin{cases} \frac{1}{2}, & 0 < t \leq 2 \\ 0. & Elsewhere \end{cases}$$

Q. 1E. Demonstrate modulation and demodulation process in Analog communication? [2][CO2][BTL3]

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Question No-2. A sinusoidal message signal

$$m(t) = 5\cos(2\pi \times 3\,000t)$$

is used to frequency modulate a carrier of frequency $f_c = 100$ MHz and frequency deviation constant $k_f = 10$ kHz/V.

(a) Determine the maximum frequency deviation Δf.
(b) Find the modulation index β.
(c) Calculate the approximate bandwidth of the FM signal using Carson's Rule. [6][CO2][BTL3]

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Question No-3. Consider $x(t) = e^{-2t}u(t)$ and $h(t) = u(t) - u(t-1)$ then Compute $y(t) = x(t) * h(t)$ where * denotes the convolution. [6][CO1][BTL2]

Question No-4. Explain in detail the Superheterodyne Receiver and support your explanation with a neat and well-labeled block diagram. [6][CO2][BTL3]

Question No-5. Write the mathematical expressions for an FM and a PM signal with a single-tone modulating signal, and highlight the similarities and differences between them. [6][CO2][BTL3]

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Question No-6. Explain in detail five major reasons why modulation is essential in electronic communication systems. Support your answer with appropriate technical justification for each reason. [8][CO1][BTL3]

Question No-7. Derive an expression for the transmission efficiency (η) of a DSB–FC (double-sideband with full carrier) amplitude modulation system when the message signal is a single tone $m(t) = M\cos(\omega_m t)$ and Carrier signal is $c(t) = A_c cos\ (\omega_c t)$. State any assumptions and notation you use in this process. Comment on the limiting cases of your expression. [8][CO2][BTL4]

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Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO1	**22**	BTL 1	**6**
		BTL 2	**8**
		BTL 3	**8**
CO2	**28**	BTL 3	**20**
		BTL 4	**8**

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